EXISTENCE OF THE DERIVED UPPER SHRIEK AS A RIGHT ADJOINT

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Allowed standard prerequisites (editorial).

- Brown representability theorem for compactly generated triangulated categories (injectives chapter, Proposition injectives-proposition-brown, canonical tag 0F5Y). cited at proof lines 4070. Provides right adjoint existence for the triangulated functor, given its compatibility with coproducts.
- The derived lower shriek Rf_! commutes with direct sums for finite type separated morphisms of quasi-compact quasi-separated schemes (more-etale, Lemma lemma-derived-lower-shriek-commute-direct-sums, canonical tag 0G29). cited at proof line 4071. The coproduct compatibility of Rf_! feeding the Brown representability argument.

Source reference. Original proposition 0F5Y: sources/stacks-chow/tags.txt, source lines 18953–18953. Cited at proof line 4070: the representability theorem yielding the right adjoint.

Source reference. Original lemma 0G29: sources/stacks-chow/tags.txt, source lines 20049–20049. Cited at proof line 4071: the coproduct compatibility hypothesis for the representability argument.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. Let Λ be a torsion coefficient ring. The functor $Rf_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ has a right adjoint $Rf^! : D(Y_{étale}, \Lambda) \to D(X_{étale}, \Lambda)$.*

Proof. This follows from Injectives, Proposition 0F5Y and Lemma 0G29 above. □

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